Section 2.9 Spheres & spatial distances

球与空间距离

Method 1 / 方法 1 [GE09-M01]

Spatial distance problems become manageable when the projection, symmetry axis or supporting plane is chosen first. A sphere center is not the same as a contact point; retain signed heights and verify containment after solving equal-distance equations.

空间距离题应先选择投影、对称轴或支撑平面。球心不是切点；解等距方程时保留有向高度，并在求解后核实容纳条件。

Sphere sections separate vertical and planar distances

球截面分离竖直距离与平面距离

Procedure

1. For sphere radius R and section radius a, write the center height as $h = \pm\sqrt{R^2 - a^2}$.

2. Decompose a center-to-center squared distance as planar distance squared plus the squared height difference.

3. When section radii are equal, subtract height squares to obtain a difference-of-squares equation.

4. After any squaring, restore the chosen side of the plane and the signs of all radicals.

通用步骤

1. 球半径为 R、截圆半径为 a 时，球心高度为 $h = \pm\sqrt{R^2 - a^2}$。

2. 把球心距平方分成平面投影距离平方与高度差平方。

3. 截圆半径相等时，相减高度平方，得到平方差方程。

4. 若曾平方，须补回球心所在半空间及各根式符号条件。

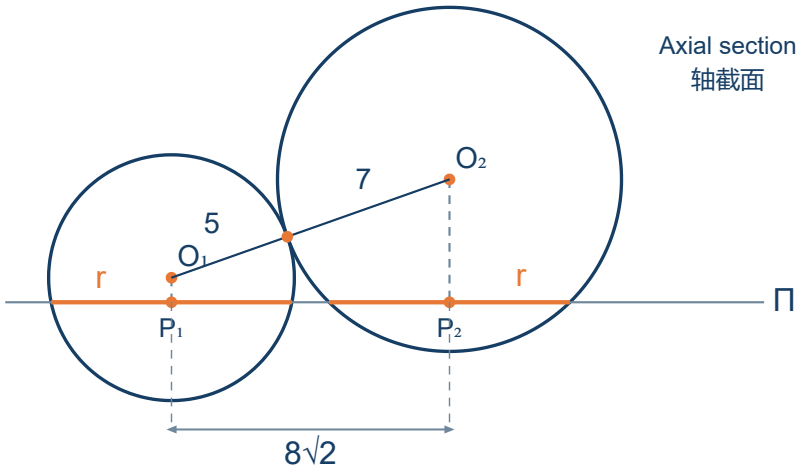
Section 2.9 Spheres & spatial distances

球与空间距离

Method 1 / 方法 1 [GE09-M01] (continued / 续)

Example. Two externally tangent spheres of radii $5, 7$ are cut into equal-radius circles by one plane. Their centers are on the same side, and their projected centers are distance $8\sqrt{2}$ apart. Find the common section radius.

例题：半径 $5, 7$ 的两外切球被同一平面截得等半径圆，球心位于平面同侧，投影圆心距为 $8\sqrt{2}$。求截圆半径。



Worked solution

1. Let positive heights be $u < v$. Since the sphere-center distance is 12, $(v-u)^2 = 144 - 128 = 16$, so $v - u = 4$.

2. Equal section radii give $v^2 - u^2 = 49 - 25 = 24$, hence $v + u = 6$ and $(u, v) = (1, 5)$.

3. The section radius squared is $25 - 1 = 24$, so it is $2\sqrt{6}$. Both heights are positive as required.

例题解析

1. 设正高度为 $u < v$。球心距为 12，故 $(v-u)^2 = 144 - 128 = 16$，得到 $v - u = 4$。

Section 2.9 Spheres & spatial distances

球与空间距离

Method 1 / 方法 1 [GE09-M01] (continued / 续)

2. 截圆半径相同给出 $v^2-u^2=49-25=24$，所以 $v+u=6$，即 $(u,v)=(1,5)$。

3. 截圆半径平方为 $25-1=24$，所以半径为 $2\sqrt{6}$；两高度均正，符合要求。

Use the difference of heights, not the difference of sphere radii.

应使用高度差，不能直接使用球半径差。

Practice / 配套练习: AIME-2020-I-Q06 (WB p. 425), AIME-2022-I-Q10 (WB p. 428)

Section 2.9 Spheres & spatial distances

球与空间距离

Method 2 / 方法 2 [GE09-M02]

Containment is a center-distance or diameter bound

容纳条件是球心距界或直径界

Procedure

1. A radius-r ball fits inside a radius-R ball precisely when the center distance is at most $R - r$.

2. For a ball resting on a plane, its center height is r; project the containment inequality to the contact plane.

3. A box needs sphere radius at least half its space diagonal because opposite vertices must both fit.

4. The box center attains that radius; for a family of boxes, maximize the diagonal separately.

通用步骤

1. 半径 r 的球能放入半径 R 的球，当且仅当球心距不超过 $R - r$。

2. 球在平面上相切时，球心高度为 r；把容纳不等式投影到接触平面。

3. 长方体需要的球半径至少是体对角线一半，因为两个对顶点均须被容纳。

4. 以长方体中心为球心可达到该半径；长方体族另需最大化体对角线。

> Example. A radius-2 ball rests on the equatorial disk of a radius-5 hemisphere and must be contained in it. What fraction of the disk allows its contact point?

> 例题：半径 2 的球在半径 5 的上半球底圆盘上相切且完全被容纳，允许接触点占底圆盘面积的几分之几?

Section 2.9 Spheres & spatial distances

球与空间距离

Method 2 / 方法 2 [GE09-M02] (continued / 续)

Worked solution

1. For contact distance d from the disk center, the small center is distance $\sqrt{d^2+4}$ from the hemisphere center.

2. Containment requires $\sqrt{d^2+4}+2 \le 5$, so $d^2 \le 5$. The small ball is automatically above the bottom plane.

3. The permitted disk has radius $\sqrt{5}$, giving area fraction $5/25 = 1/5$.

例题解析

1. 设接触点距底面圆心为 d，小球心距半球心为 $\sqrt{d^2+4}$。

2. 容纳要求 $\sqrt{d^2+4}+2 \le 5$，即 $d^2 \le 5$；小球同时自动位于底面上方。

3. 允许圆盘半径为 $\sqrt{5}$，面积占比为 $5/25 = 1/5$。

The containing sphere can be positioned separately for each box; its radius is the common worst-case requirement.

对每个长方体可分别摆放容球；共同要求的是最坏情形半径。

Practice / 配套练习: AIME-2024-I-Q15 (WB p. 189), AIME-2026-I-Q03 (WB p. 424)

Section 2.9 Spheres & spatial distances

球与空间距离

Method 3 / 方法 3 [GE09-M03]

Use normalized distances to supporting planes

使用归一化支撑平面距离

Procedure

1. Write each face or support as $ax+by+cz=d$ and divide signed distance by $\sqrt{a^2+b^2+c^2}$.

2. Choose an interior sign convention before equating face distances to a radius.

3. Exploit coordinate-plane tangency or reflection symmetry to reduce center variables.

4. For a largest insphere, prove a bound using supporting planes and then verify one center attains all inequalities.

通用步骤

1. 将每个面或支撑写为 $ax+by+cz=d$，有向距离须除以 $\sqrt{a^2+b^2+c^2}$。

2. 令各面距离等于半径前，先固定内部的正负号约定。

3. 利用坐标面相切或反射对称减少球心变量。

4. 求最大内球时，先由支撑平面证明界，再核实某球心满足全部不等式并取等。

> Example. Find the inradius of the tetrahedron in the first octant bounded by $x+y+z=6$.

> 例题：求第一卦限内由 $x+y+z=6$ 与三个坐标面围成的四面体内切球半径。

Section 2.9 Spheres & spatial distances

球与空间距离

Method 3 / 方法 3 [GE09-M03] (continued / 续)

Worked solution

1. Tangency to the coordinate planes gives center (r, r, r) with $r > 0$.

2. The distance to the fourth face is $(6 - 3r)/\sqrt{3}$, so $6 - 3r = r\sqrt{3}$.

3. Hence $r = 6/(3 + \sqrt{3}) = 3 - \sqrt{3}$. Its center is interior and the four positive face distances all equal r.

例题解析

1. 与三个坐标面相切使球心为 (r, r, r)，其中 $r > 0$。

2. 到第四面的距离为 $(6 - 3r)/\sqrt{3}$，故 $6 - 3r = r\sqrt{3}$。

3. 所以 $r = 3 - \sqrt{3}$。球心在内部，到四面的正距离均为 r。

Plane coefficients alone are not distances: normalization matters.

平面系数不是距离，法向量长度的归一化不可省略。

For a solid cone described by $\alpha u + \beta\rho \le \gamma$, where u is the axial coordinate, ρ the radial coordinate and $\beta > 0$, every lateral supporting plane stays at least a ball radius away precisely when $\sqrt{\alpha^2 + \beta^2}\, r \le \gamma - \alpha u - \beta\rho$ at its center. Two perpendicular cones give coupled absolute-value bounds; add them and check the base-plane distances too.

A rigid tripod can be kept fixed while its three contact points define a new support plane. The height is the distance of the apex to that plane, avoiding a full spatial rotation.

Section 2.9 Spheres & spatial distances

球与空间距离

Method 3 / 方法 3 [GE09-M03] (continued / 续)

若实心圆锥由 $\alpha u + \beta\rho \le \gamma$ 描述，其中 u 为轴向坐标、ρ 为径向坐标且 $\beta > 0$，则各侧面支撑平面到球心距离均不小于球半径，等价于球心满足 $\sqrt{\alpha^2 + \beta^2}\, r \le \gamma - \alpha u - \beta\rho$。两个垂直圆锥给出含绝对值的耦合界，可相加求界，但仍需检验底面距离。

刚性三脚架可保持自身坐标不变，由三个接触点确定新支撑平面；高度即顶点到该平面距离，无需求整体空间旋转。

Practice / 配套练习: AIME-2001-I-Q12 (WB p. 431), AIME-2006-I-Q14 (WB p. 393), AIME-2020-II-Q07 (WB p. 426), AIME-2026-II-Q12 (WB p. 435)

Section 2.9 Spheres & spatial distances

球与空间距离

Method 4 / 方法 4 [GE09-M04]

Symmetry locates an axial circumsphere center

对称性确定轴上的外接球心

Procedure

1. Project a sphere center to a plane through three sphere points; the projection is their planar circumcenter.

2. For an axial pyramid or symmetric tetrahedron, place the center at signed height z on the symmetry axis.

3. Equate squared distances to an apex and a base vertex; quadratic terms usually cancel.

4. When asked for distance to a filled triangle, verify the perpendicular foot is inside that triangle.

通用步骤

1. 球心向三个球面点所在平面投影，垂足即三点的平面外心。

2. 对轴对称棱锥或对称四面体，把球心放在对称轴有向高度 z 处。

3. 令到顶点和底面顶点的距离平方相等，二次项通常会消去。

4. 若求到填满三角形的距离，必须确认垂足在三角形内部。

> Example. A right square pyramid has base edge 4 and height 3. Find the radius of its circumsphere.

> 例题：正四棱锥底边长 4、高 3，求外接球半径。

Section 2.9 Spheres & spatial distances

球与空间距离

Method 4 / 方法 4 [GE09-M04] (continued / 续)

Worked solution

1. Put the base center at height zero and let the sphere center have signed height z. A base vertex is planar distance $2\sqrt{2}$ from the axis.

2. Equate $z^2 + 8 = (3 - z)^2$, obtaining $z = 1/6$.

3. The radius is $3 - z = 17/6$. Equal distances to all four base vertices follow by symmetry.

例题解析

1. 以底面中心为零高度，球心有向高度为 z；底顶点到轴的平面距离为 $2\sqrt{2}$。

2. 由 $z^2 + 8 = (3 - z)^2$ 得 $z = 1/6$。

3. 半径为 $3 - z = 17/6$，对称性保证到另三个底顶点距离相同。

The center can lie below a base plane; do not assume $z > 0$ without proof.

球心可能位于底面下方，不可未经证明假设 $z > 0$。

For a triangle of circumradius R on a sphere of radius L, its plane distance is $\sqrt{L^2 - R^2}$. This equals distance to the filled triangle only when its circumcenter lies inside, for example when the triangle is acute.

若球面上三角形外接圆半径为 R、球半径为 L，球心到其平面的距离为 $\sqrt{L^2 - R^2}$。只有外心在三角形内部时，才等于到填满三角形的距离；锐角三角形即满足此条件。

Practice / 配套练习: AIME-2000-II-Q12 (WB p. 430), AIME-2022-II-Q03 (WB p. 423), AIME-2026-II-Q12 (WB p. 435)

Section 2.9 Spheres & spatial distances

球与空间距离

Method 5 / 方法 5 [GE09-M05]

Orthonormal edge projections have a fixed square sum

正交单位边投影具有固定平方和

Procedure

1. Use three mutually perpendicular edge directions as an orthonormal basis.

2. The components of a unit vertical vector in that basis have squared sum one.

3. Multiply by edge length squared to relate three height differences.

4. Choose the root that gives all required projection signs and the specified nearest or farthest vertex.

通用步骤

1. 以三条相互垂直的棱方向作为标准正交基。

2. 竖直单位向量在此基中的三个分量平方和为一。

3. 乘以棱长平方，得到三个高度差的关系。

4. 根据投影正负及最近、最远顶点要求筛选代数根。

> Example. A side-3 cube has a lowest vertex at height $h > 0$, and its three adjacent vertices all at height 2. Find h.

> 例题：棱长 3 的立方体最低顶点高度为 $h > 0$，三个相邻顶点高度都为 2，求 h。

Worked solution

1. The three edge height projections are all $2 - h > 0$. Their squared sum must equal 3^2.

Section 2.9 Spheres & spatial distances

球与空间距离

Method 5 / 方法 5 [GE09-M05] (continued / 续)

2. Thus $3(2-h)^2 = 9$, giving $h = 2 \pm \sqrt{3}$.

3. The plus sign makes adjacent vertices lower, contradicting the premise. Hence $h = 2 - \sqrt{3} > 0$.

例题解析

1. 三条棱的高度投影均为正数 $2-h$，平方和必须为 3^2。

2. 所以 $3(2-h)^2 = 9$，得 $h = 2 \pm \sqrt{3}$。

3. 正号会使相邻顶点更低，故只能取 $h = 2 - \sqrt{3} > 0$。

The fixed square sum comes from orthogonality, not equal inclination.

固定平方和来自正交性，不依赖三条棱倾角相同。

Practice / 配套练习: AIME-2011-I-Q13 (WB p. 432)

Section 2.9 Spheres & spatial distances

球与空间距离

Method 6 / 方法 6 [GE09-M06]

A perpendicular section converts dihedrals into tangent-center triangles

垂直截面将二面角化为切球圆心三角形

Procedure

1. Cut perpendicularly to the intersection line of the planes; their dihedral becomes a planar angle.

2. A sphere tangent to both planes has its center on an angle bisector, transverse distance $r / \sin\theta$ for half-angle θ.

3. Keep the along-line coordinate separate and use tangency center distances in the resulting right triangle.

4. For axial sphere points, identify an actual diameter before using a right-angle or altitude-product identity.

通用步骤

1. 作垂直于两平面交线的截面，将二面角转为平面角。

2. 同时与两面相切的球心在角平分线上，半角为 θ 时横向距为 $r / \sin\theta$。

3. 交线方向坐标另行保留，在所得直角三角形中使用切球心距。

4. 对轴上球面点，先证实为直径端点，再使用直角或高线乘积关系。

> Example. Two perpendicular planes meet along a line. Two spheres of radius 2 lie in the same sector, tangent to both planes and each other. What is the separation of their centers along the intersection line?

Section 2.9 Spheres & spatial distances

球与空间距离

Method 6 / 方法 6 [GE09-M06] (continued / 续)

> 例题：两垂直平面相交于一直线，两个半径 2 的球位于同一角域，同时与两平面及彼此外切。求两球心沿交线方向的距离。

Worked solution

1. In the perpendicular cross section, both centers have coordinates $(2, 2)$ relative to the planes, so their transverse projections coincide.

2. External tangency makes their full center distance $2 + 2 = 4$.

3. All of that distance is along the intersection line, so the requested separation is 4.

例题解析

1. 在垂直截面内，两球心相对两面的坐标均为 $(2, 2)$，所以横向投影相同。

2. 外切使球心距为 $2 + 2 = 4$。

3. 该距离全部沿交线方向，故所求为 4。

The two-plane cross section does not capture separation parallel to their intersection line.

两平面的垂直截面不会显示沿交线方向的间距。

For unequal radii R, r in the same sector, the transverse center separation is $|R - r| / \sin\theta$. If a small sphere touches two equal spheres whose centers differ only along the intersection line, it lies halfway between them along that line.

Section 2.9 Spheres & spatial distances

球与空间距离

Method 6 / 方法 6 [GE09-M06] (continued / 续)

A sphere through three noncollinear triangle vertices has its center on their perpendicular circumcenter axis. Two distinct sphere points on that axis are diameter endpoints. A triangle vertex then supplies a right triangle and, when its projection lies between the endpoints, the altitude-product relation.

同一角域内半径为 R, r 的球心横向间距为 $|R-r|/\sin\theta$。若小球同时与两个仅沿交线分离的等球相切，则小球在交线方向上位于两者正中。

经过三角形三个不共线顶点的球，其球心在外心垂轴上。该轴上两个不同球面点必为直径端点。再用一个三角形顶点构造直角三角形；若其投影位于两端点之间，还可使用高线乘积关系。

Practice / 配套练习: AIME-2016-II-Q14 (WB p. 433), AIME-2021-II-Q10 (WB p. 427)

Section 2.9 Spheres & spatial distances

球与空间距离

Method 7 / 方法 7 [GE09-M07]

Prism distances split into horizontal chord and height classes

棱柱距离分成水平弦长与高度类别

Procedure

1. Classify pairs by their horizontal separation in the base polygon.

2. For vertices in different parallel bases, add the fixed height square to the horizontal distance square.

3. Test each possible equality of two sides, including equality with a base side rather than only two cross-base sides.

4. Count disjoint vertex classes and multiply by symmetry only when it preserves the class.

通用步骤

1. 按底面多边形中的水平距离对点对分类。

2. 不同平行底面的顶点距离平方，等于水平距离平方加固定高度平方。

3. 检查各种两边相等条件，既包括两条跨底边相等，也包括跨底边与底边相等。

4. 对互斥顶点类别计数，仅在对称保持类别时使用对称倍数。

> Example. A right square prism has unit base edge and height $\sqrt{2}$. Choose a diagonal pair in its bottom base. How many top vertices form an isosceles triangle with that pair?

> 例题：正四棱柱底边为一、高为 $\sqrt{2}$。固定底面一对对角顶点，顶面有多少顶点能与之组成等腰三角形?

Section 2.9 Spheres & spatial distances

球与空间距离

Method 7 / 方法 7 [GE09-M07] (continued / 续)

Worked solution

1. The bottom diagonal has length $\sqrt{2}$. A top vertex above either endpoint has cross distances $\sqrt{2}, 2$, so it forms an isosceles triangle by matching the bottom diagonal.

2. Each of the other two top vertices has horizontal distance one to both bottom endpoints, hence both cross distances are $\sqrt{3}$.

3. All four top vertices qualify. The two mechanisms are disjoint and neither may be omitted.

例题解析

1. 底面对角线长为 $\sqrt{2}$。正上方对应两个端点的顶点，其跨底距离为 $\sqrt{2}, 2$，通过底边与跨底边相等构成等腰。

2. 另外两个顶点到两底端点的水平距离均为一，跨底距离均为 $\sqrt{3}$。

3. 四个顶点全部符合；两种机制互斥且不能漏掉任一种。

Equal-height symmetry is only one way to make an isosceles spatial triangle.

等高对称不是空间等腰三角形的唯一来源。

Practice / 配套练习: AIME-2018-I-Q07 (WB p. 699)

Section 2.9 Spheres & spatial distances

球与空间距离

Method 8 / 方法 8 [GE09-M08]

An axial torus section gives the contact-circle radius

圆环面轴截面给出接触圆半径

Procedure

1. In the coaxial configuration, the sphere center lies on the torus rotation axis. A plane through that axis cuts the torus into its generating circles; analyze either one.

2. For sphere radius R and tube radius r, use center distance $R - r$ for internal tangency (requiring $R > r$), or $R + r$ for external tangency.

3. The contact point lies on the center line, so its distance to the axis scales by $R/(R \mp r)$.

4. Check that the required center distance is at least the major torus radius, so the axial placement exists.

通用步骤

1. 在同轴构型中，球心位于圆环旋转轴上。经过该轴的平面将圆环面截成母圆，分析其中一个即可。

2. 球半径为 R、管半径为 r 时，内切圆心距为 $R - r$（要求 $R > r$），外切为 $R + r$。

3. 切点在连心线上，到轴距离按 $R/(R \mp r)$ 缩放。

4. 检查所需圆心距不小于圆环主半径，保证轴向摆放存在。

> Example. A torus has major radius 3 and tube radius 1. Find its contact-circle radius when internally tangent to a sphere of radius 5.

> 例题：圆环面主半径为 3、管半径为 1，与半径 5 的球从内部相切，求接触圆半径。

Section 2.9 Spheres & spatial distances

球与空间距离

Method 8 / 方法 8 [GE09-M08] (continued / 续)

Worked solution

1. The generating-circle center is distance $5-1=4$ from the sphere center and distance 3 from the axis.

2. An axial displacement $\sqrt{4^2-3^2}=\sqrt{7}$ exists, so this placement is feasible.

3. The contact point is $5/4$ times as far from the axis as the generating-circle center. The contact-circle radius is $15/4$.

例题解析

1. 母圆圆心距球心为 $5-1=4$，距旋转轴为 3。

2. 轴向位移为 $\sqrt{4^2-3^2}=\sqrt{7}$，所以摆放可行。

3. 切点到轴距离是母圆心的 $5/4$ 倍，故接触圆半径为 $15/4$。

The contact circle is not generally the torus midline circle.

接触圆通常不是圆环的中心圆。

Practice / 配套练习: AIME-2024-II-Q08 (WB p. 429)

Section 2.9 Spheres & spatial distances

球与空间距离

Method 9 / 方法 9 [GE09-M09]

Two common tangent planes make sphere radii affine

两个公共切平面使球半径成为仿射函数

Procedure

1. Put one tangent plane at $z = 0$, giving center (x, y, ρ) for radius ρ.

2. Write the second plane with unit normal (u, v, h) and a consistent signed-distance equation.

3. Its tangency equation is linear in x, y, ρ; for a distinct plane solve $\rho = \alpha x + \beta y + \gamma$.

4. Interpolate this affine function from three noncollinear contact points and verify a unit normal exists.

通用步骤

1. 把一个切平面设为 $z = 0$，半径 ρ 的球心即 (x, y, ρ)。

2. 以单位法向量 (u, v, h) 表示第二平面，保持有向距离符号一致。

3. 切触式对 x, y, ρ 线性；平面不同便可解为 $\rho = \alpha x + \beta y + \gamma$。

4. 用三个不共线接触点插值仿射函数，并验证存在单位法向量。

> Example. Spheres tangent above $z = 0$ have radii $1, 2, 3$ at contact points $(0, 0), (2, 0), (0, 2)$. They share a second tangent plane on the same side. Find the radius at contact point $(1, 1)$ that shares this plane.

> 例题：在 $z = 0$ 上方相切的球，在接触点 $(0, 0), (2, 0), (0, 2)$ 处半径分别为 $1, 2, 3$，并与另一个平面同侧相切。求接触点 $(1, 1)$ 处与该平面也相切的球半径。

Section 2.9 Spheres & spatial distances

球与空间距离

Method 9 / 方法 9 [GE09-M09] (continued / 续)

Worked solution

1. The interpolating radius function is $g(x, y) = 1 + x/2 + y$. Therefore the desired radius is $g(1, 1) = 5/2$.

2. To verify the plane, set $u = (1 - h)/2, v = 1 - h$; the unit-normal equation gives $h = 1$ or $h = 1/9$.

3. Take the distinct-plane value $h = 1/9$, yielding $(u, v, h) = (4, 8, 1)/9$ and plane $4x + 8y + z = -8$. Its signed distance from each center is exactly its radius.

例题解析

1. 插值得半径函数 $g(x, y) = 1 + x/2 + y$，因此所求为 $5/2$。

2. 验证平面时，取 $u = (1 - h)/2, v = 1 - h$；单位法向条件给出 $h = 1$ 或 $h = 1/9$。

3. 取不同平面对应的 $h = 1/9$，得单位法向量 $(4, 8, 1)/9$ 及平面 $4x + 8y + z = -8$。各球心到该面的有向距离恰为其半径。

The $h = 1$ solution reproduces the first plane and must be excluded.

$h = 1$ 对应原来的第一个平面，必须排除。

Practice / 配套练习: AIME-2026-I-Q12 (WB p. 434)